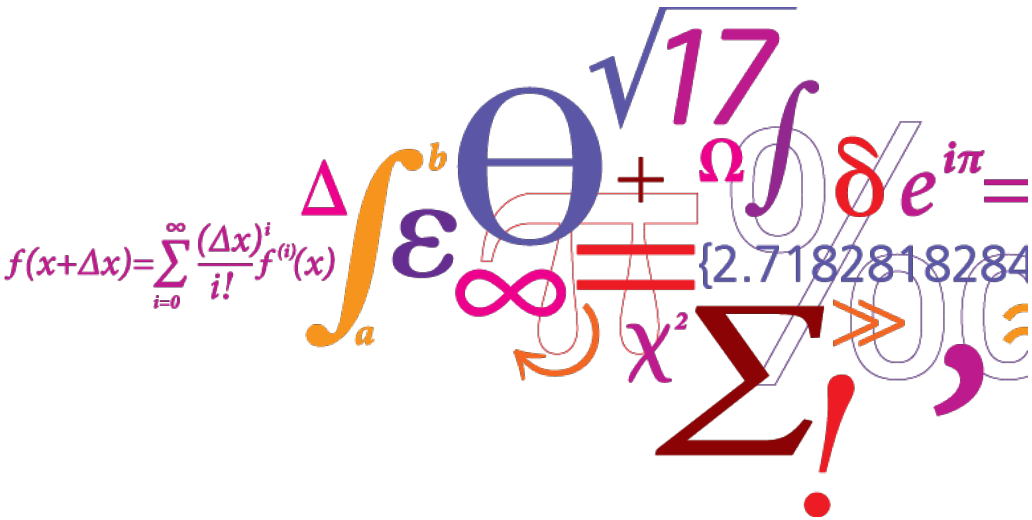
31761 – Renewables in Electricity Markets

Lecture 12b: Incorporation of unit commitment constraints

Jalal Kazempour



May 2, 2022

DTU Electrical Engineering
Department of Electrical Engineering

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Ramp-up constraints

$$p_{g,t}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \leq R_g^{\mathrm{UP}} \qquad \forall g, t > 1$$

$$p_{g,t}^{\mathrm{G}} - p_g^{\mathrm{G,initial}} \leq R_g^{\mathrm{UP}} \qquad \forall g, t = 1$$

Ramp-down constraints

$$p_{g,t}^{\mathrm{G}} - p_{g,t-1}^{\mathrm{G}} \geq -R_g^{\mathrm{DN}} \qquad \forall g, t > 1$$

$$p_{g,t}^{\mathrm{G}} - p_g^{\mathrm{G,initial}} \geq -R_g^{\mathrm{DN}} \qquad \forall g, t = 1$$

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Ramp-up constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \leq R^{\mathrm{UP}}_{g} \qquad \forall g, \boxed{t} > 1$$

Index for time periods

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \leq R^{\mathrm{UP}}_{g} \qquad \forall g, t = 1$$

Ramp-down constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t > 1$$

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t = 1$$

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Ramp-up constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \leq R^{\mathrm{UP}}_{g} \qquad \forall \boxed{g}, t > 1$$

Index for conventional generators

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \leq R^{\mathrm{UP}}_{g} \qquad \forall g, t = 1$$

Ramp-down constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t > 1$$

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t = 1$$

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Ramp-up constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \leq \boxed{R^{\mathrm{UP}}_g} \quad \forall g, t > 1$$

Ramp-up limit (MW/hour), given value

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \leq R^{\mathrm{UP}}_g \quad \forall g, t = 1$$

Ramp-down constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \geq -\boxed{R^{\mathrm{DN}}_g} \quad \forall g, t > 1$$

Ramp-down limit (MW/hour), given value

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \geq -R^{\mathrm{DN}}_g \quad \forall g, t = 1$$

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Ramp-up constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \leq R^{\mathrm{UP}}_{g} \qquad \forall g, t > 1$$

$$p^{\mathrm{G}}_{g,t} - \boxed{p^{\mathrm{G,initial}}_{g}} \leq R^{\mathrm{UP}}_{g} \qquad \forall g, t = 1$$

Production of generator *g* in the beginning of time horizon (MW), given value

Ramp-down constraints

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G}}_{g,t-1} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t > 1$$

$$p^{\mathrm{G}}_{g,t} - p^{\mathrm{G,initial}}_{g} \geq -R^{\mathrm{DN}}_{g} \qquad \forall g, t = 1$$

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

Unit commitment

In the previous lectures, we have ignored unit commitment (UC) constraints of conventional generators within the market-clearing problem.

These UC constraints include:

- Ramping limits of generators,
- The minimum production level of generators,
- Start-up cost of generators,
- Minimum up- and down-time constraints of generators,
- etc.

These constraints require binary (0/1) variables!

Discussion

Note: The unit commitment problem is a mixed-integer linear problem (MILP), and we still need dual variables to derive market prices!

Discussion

Note: The unit commitment problem is a mixed-integer linear problem (MILP), and we still need dual variables to derive market prices!

Question: How to derive dual variables in a MILP? Is there any mathematical challenge?

Discussion

Note: The unit commitment problem is a mixed-integer linear problem (MILP), and we still need dual variables to derive market prices!

Question: How to derive dual variables in a MILP? Is there any mathematical challenge?

Questions: When UC constraints are included,

How to derive “market prices”?

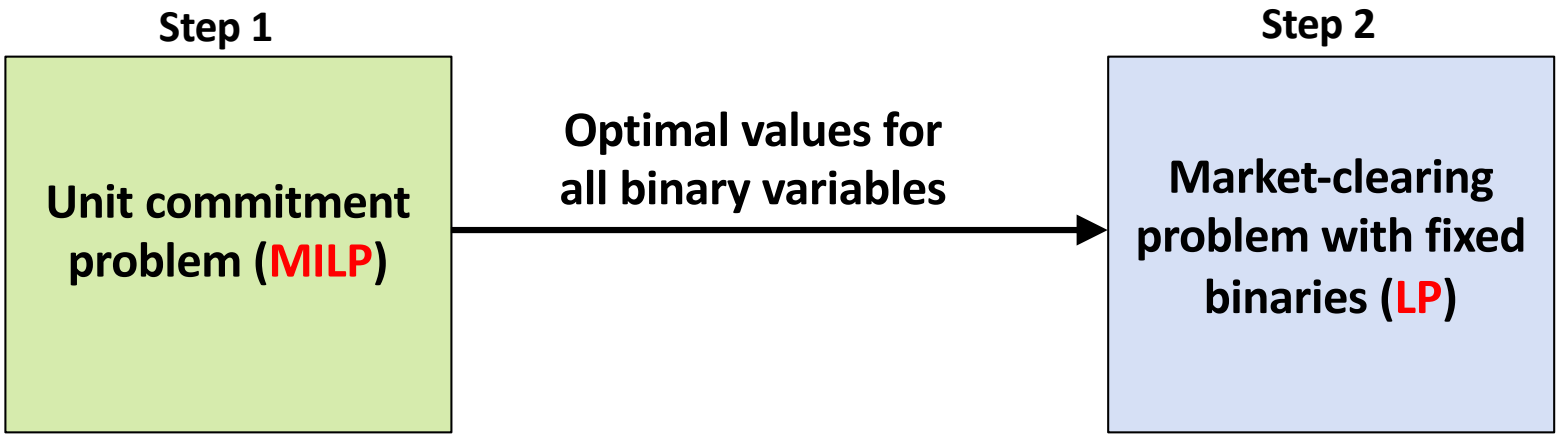
Pricing with unit commitment constraints

Step 1

Unit commitment problem (MILP)

- **Step 1:** Solve the unit commitment problem, and obtain the optimal values for all binary variables

Pricing with unit commitment constraints



- **Step 1:** Solve the unit commitment problem, and obtain the optimal values for all binary variables

- **Step 2:** Solve the same problem while binary variables are replaced by their optimal values (0 or 1) obtained in Step 1. This results in a linear problem (LP), so determine the market prices based on dual variables.

Step 1: unit commitment problem

Step 1: unit commitment problem

Let's add the following UC constraints to the market-clearing problem:

- The minimum production level of generators,
- Start-up cost of generators.

Step 1: unit commitment problem

$$\underset{p^{\mathrm{G}}_{g,t}, u^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [s^{\mathrm{G}}_{g,t} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G}}_{g,t} - u^{\mathrm{G}}_{g,t-1}) \quad \forall g, \forall t$$

$$u^{\mathrm{G}}_{g,t} \in \{0, 1\} \quad \forall g, \forall t$$

Step 1: unit commitment problem

$$\underset{p^{\mathrm{G}}_{g,t}, u^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [s^{\mathrm{G}}_{g,t} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \boxed{\forall t}$$ Index for time periods

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G}}_{g,t} - u^{\mathrm{G}}_{g,t-1}) \quad \forall g, \forall t$$

$$u^{\mathrm{G}}_{g,t} \in \{0, 1\} \quad \forall g, \forall t$$

Step 1: unit commitment problem

$$\underset{p^{\mathrm{G}}_{g,t}, u^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [s^{\mathrm{G}}_{g,t} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \boxed{\forall t}$$ Index for time periods

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G}}_{g,t} - u^{\mathrm{G}}_{g,t-1}) \quad \forall g, \forall t$$

Binary variables $\boxed{u^{\mathrm{G}}_{g,t}} \in \{0, 1\} \quad \forall g, \forall t$

Step 1: unit commitment problem

$$\underset{p^{G}_{g,t}, u^{G}_{g,t}, s^{G}_{g,t},\ p^{D}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{D}_{d,t} - \sum_{g,t} [s^{G}_{g,t} + (C_g\ p^{G}_{g,t})]$$

subject to:

$$0 \le p^{D}_{d,t} \le \overline{P}^{D}_{d,t} \quad \forall d, \boxed{\forall t}$$

Index for time periods

Binary variable appears in the lower and upper bounds of generation level

$$\boxed{\underline{P}^{G}_{g,t}\ u^{G}_{g,t}} \le p^{G}_{g,t} \le \boxed{\overline{P}^{G}_{g}\ u^{G}_{g,t}} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{D}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{G}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \le s^{G}_{g,t} \ge C^{SU}_{g}(u^{G}_{g,t} - u^{G}_{g,t-1}) \quad \forall g, \forall t$$

Binary variables

$$\boxed{u^{G}_{g,t}} \in \{0, 1\} \quad \forall g, \forall t$$

Step 1: unit commitment problem

$$\underset{p^{G}_{g,t}, u^{G}_{g,t}, s^{G}_{g,t},\ p^{D}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{D}_{d,t} - \sum_{g,t} [\boxed{s^{G}_{g,t}} + (C_g\ p^{G}_{g,t})]$$

Variable: start-up cost of generator *g* in time *t*

subject to:

$$0 \le p^{D}_{d,t} \le \overline{P}^{D}_{d,t} \quad \forall d, \boxed{\forall t}$$

Index for time periods

Binary variable appears in the lower and upper bounds of generation level

$$\boxed{\underline{P}^{G}_{g,t}\ u^{G}_{g,t}} \le p^{G}_{g,t} \le \boxed{\overline{P}^{G}_{g}\ u^{G}_{g,t}} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{D}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{G}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \le s^{G}_{g,t} \ge C^{SU}_{g}(u^{G}_{g,t} - u^{G}_{g,t-1}) \quad \forall g, \forall t$$

Binary variables

$$\boxed{u^{G}_{g,t}} \in \{0, 1\} \quad \forall g, \forall t$$

Step 1: unit commitment problem

$$\underset{p^{G}_{g,t}, u^{G}_{g,t}, s^{G}_{g,t},\ p^{D}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{D}_{d,t} - \sum_{g,t} [s^{G}_{g,t} + (C_g\ p^{G}_{g,t})]$$

Variable: start-up cost of generator *g* in time *t*

subject to:

$$0 \le p^{D}_{d,t} \le \overline{P}^{D}_{d,t} \quad \forall d, \forall t$$

Index for time periods

Binary variable appears in the lower and upper bounds of generation level

$$\underline{P}^{G}_{g,t}\ u^{G}_{g,t} \le p^{G}_{g,t} \le \overline{P}^{G}_{g}\ u^{G}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{D}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{G}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \le s^{G}_{g,t} \ge C^{SU}_{g}(u^{G}_{g,t} - u^{G}_{g,t-1}) \quad \forall g, \forall t$$

Start-up cost constraints

Binary variables

$$u^{G}_{g,t} \in \{0, 1\} \quad \forall g, \forall t$$

Step 1: unit commitment problem

$$\underset{p^{G}_{g,t}, u^{G}_{g,t}, s^{G}_{g,t},\ p^{D}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{D}_{d,t} - \sum_{g,t} [s^{G}_{g,t} + (C_g\ p^{G}_{g,t})]$$

Variable: start-up cost of generator *g* in time *t*

subject to:

$$0 \le p^{D}_{d,t} \le \overline{P}^{D}_{d,t} \quad \forall d, \forall t$$

Index for time periods

Binary variable appears in the lower and upper bounds of generation level

$$\underline{P}^{G}_{g,t}\ u^{G}_{g,t} \le p^{G}_{g,t} \le \overline{P}^{G}_{g}\ u^{G}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{D}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{G}_{g,t} = 0 \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

Parameter: start-up cost of generator *g*

$$0 \le s^{G}_{g,t} \ge C^{SU}_{g}(u^{G}_{g,t} - u^{G}_{g,t-1}) \quad \forall g, \forall t$$

Start-up cost constraints

Binary variables

$$u^{G}_{g,t} \in \{0, 1\} \quad \forall g, \forall t$$

Step 2: LP with fixed binaries

Step 2: LP with fixed binaries

$$\underset{p^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [s^{\mathrm{G}}_{g,t} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G,fixed}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G,fixed}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad : \lambda_{n,t} \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G,fixed}}_{g,t} - u^{\mathrm{G,fixed}}_{g,t-1}) \quad \forall g, \forall t$$

Step 2: LP with fixed binaries

$$\underset{p^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [\boxed{s^{\mathrm{G}}_{g,t}} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G,fixed}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G,fixed}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad : \boxed{\lambda_{n,t}} \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G,fixed}}_{g,t} - u^{\mathrm{G,fixed}}_{g,t-1}) \quad \forall g, \forall t$$

Questions:
Does the start-up cost affect market prices?

Step 2: LP with fixed binaries

$$\underset{p^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [\boxed{s^{\mathrm{G}}_{g,t}} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!

subject to:

$$0 \le p^{\mathrm{D}}_{d,t} \le \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G,fixed}}_{g,t} \le p^{\mathrm{G}}_{g,t} \le \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G,fixed}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad : \boxed{\lambda_{n,t}} \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \le s^{\mathrm{G}}_{g,t} \ge C^{\mathrm{SU}}_{g}(u^{\mathrm{G,fixed}}_{g,t} - u^{\mathrm{G,fixed}}_{g,t-1}) \quad \forall g, \forall t$$

Step 2: LP with fixed binaries

$$\underset{p^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [\boxed{s^{\mathrm{G}}_{g,t}} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \leq p^{\mathrm{D}}_{d,t} \leq \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G,fixed}}_{g,t} \leq p^{\mathrm{G}}_{g,t} \leq \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G,fixed}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad : \boxed{\lambda_{n,t}} \quad \forall n, \forall t$$

$$-F_{n,m} \leq B_{n,m}(\theta_{n,t} - \theta_{m,t}) \leq F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\theta_{(n=ref),t} = 0 \quad \forall t$$

$$0 \leq s^{\mathrm{G}}_{g,t} \geq C^{\mathrm{SU}}_{g}(u^{\mathrm{G,fixed}}_{g,t} - u^{\mathrm{G,fixed}}_{g,t-1}) \quad \forall g, \forall t$$

By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!

The start-up cost does NOT contribute to the market price formation, because that cost is not reflected in the dual variable!

Step 2: LP with fixed binaries

$$\underset{p^{\mathrm{G}}_{g,t}, s^{\mathrm{G}}_{g,t},\ p^{\mathrm{D}}_{d,t},\ \theta_{n,t}}{\text{Maximize}} \quad \sum_{d,t} U_{d,t}\ p^{\mathrm{D}}_{d,t} - \sum_{g,t} [\boxed{s^{\mathrm{G}}_{g,t}} + (C_g\ p^{\mathrm{G}}_{g,t})]$$

subject to:

$$0 \le p^{\mathrm{D}}_{d,t} \le \overline{P}^{\mathrm{D}}_{d,t} \quad \forall d, \forall t$$

$$\underline{P}^{\mathrm{G}}_{g,t}\ u^{\mathrm{G,fixed}}_{g,t} \le p^{\mathrm{G}}_{g,t} \le \overline{P}^{\mathrm{G}}_{g}\ u^{\mathrm{G,fixed}}_{g,t} \quad \forall g, \forall t$$

$$\sum_{d \in \Psi_n} p^{\mathrm{D}}_{d,t} + \sum_{m \in \Omega_n} B_{n,m}(\theta_{n,t} - \theta_{m,t}) - \sum_{g \in \Psi_n} p^{\mathrm{G}}_{g,t} = 0 \quad : \boxed{\lambda_{n,t}} \quad \forall n, \forall t$$

$$-F_{n,m} \le B_{n,m}(\theta_{n,t} - \theta_{m,t}) \le F_{n,m} \quad \forall n, \forall m \in \Omega_n, \forall t$$

$$\forall g, \forall t$$

By fixing the values of binaries, the optimal values of the start-up cost of generators are somehow given! Their start-up cost variable looks like a parameter in the objective function, and can be eliminated!

The start-up cost does NOT contribute to the market price formation, because that cost is not reflected in the dual variable!

This means that the "cost recovery" of generators **cannot** be ensured anymore (i.e., their profit may become negative), because market prices do not support their start-up costs!

Step 2: LP with fixed binaries

- In current practice of electricity markets in the U.S., market operators use "**uplift mechanisms**" to restore "cost recovery", but at the cost of market inefficiency and sub-optimality!
- The negative profit of each generator is compensated by an "ex-post" side payment, i.e., "uplift payment".
- The loads will be eventually charged due to uplift payments!

Step 2: LP with fixed binaries

- In current practice of electricity markets in the U.S., market operators use "**uplift mechanisms**" to restore "cost recovery", but at the cost of market inefficiency and sub-optimality!
- The negative profit of each generator is compensated by an "ex-post" side payment, i.e., "uplift payment".
- The loads will be eventually charged due to uplift payments!

Discussion:
What is the current practice in the European electricity markets?
Guide: Read about "block orders" and "paradoxically rejected blocks"

Thanks for your attention!

Email: seykaz@dtu.dk